Study Skills for Science: Notes, Examples and Key Terms

Estimating Rates of Change from Tables of Measured Values Collected During Class Experiments

A table of values is often the first form in which a rate of change is met. The difference quotient between neighbouring rows is the best guide to the behaviour of the underlying quantity.

Check the units before you start. Convert every quantity to a single system of units before substituting any numbers.

Why does the sign change? Dividing both sides of an inequality by a negative number reverses its direction.

Notation Vectors are written in bold type and scalars in italics.

A short Python function for this step is shown below.

```
def bisect(f, lo, hi, tol=1e-9):
    # f(lo) and f(hi) must have opposite signs
    while hi - lo > tol:
        mid = (lo + hi) / 2
        if f(lo) * f(mid) <= 0:
            hi = mid
        else:
            lo = mid
    return (lo + hi) / 2
```

A short Python function for this step is shown below.

```
def trapezoid(f, a, b, n=100):
    h = (b - a) / n
    total = (f(a) + f(b)) / 2
    for k in range(1, n):
        total += f(a + k * h)
    return total * h
```

Reading Graphs of Velocity Against Time and Recovering the Distance Travelled by an Object

The area between a velocity graph and the time axis equals the displacement. Signs matter: regions below the axis count as motion in the opposite direction.

Remember the order of operations: Brackets first, then powers, then multiplication and division, and last addition and subtraction.

Do not round early! Carry at least two more digits than you need until the last line of the calculation.

Caution Calculators may be set to degrees or to radians; check before using trigonometric functions.

Reading list. Chapter 3 of the course notes, the practice sheet on rates of change, and the short article on measurement error.

Schedule. Lecture on Monday, laboratory on Wednesday, and a quiz at the start of class on Friday morning.

Example

A cart rolls 12 m down a ramp in 4.0 s, starting from rest. Find its acceleration.

Solution

Use d = (1/2) a t^2, so a = 2d / t^2 = 2(12) / $(4.0)^2$ = 1.5 m/s^2.

A short Python function for this step is shown below.

```
def newton(f, df, x, steps=8):
    for _ in range(steps):
        x = x - f(x) / df(x)
    return x

print(newton(lambda t: t * t - 2, lambda t: 2 * t, 1.0))
```

A short Python function for this step is shown below.

```
def gcd(a, b):
    while b:
        a, b = b, a % b
    return a

print(gcd(84, 36))
```

How Rounding Errors Accumulate When a Long Calculation Is Carried Out on a Pocket Calculator

Every rounded intermediate result introduces a small error, and these errors can grow. Keeping extra digits until the final step is a simple habit that avoids most of the damage.

Write down what you know. A short list of given quantities and unknowns often shows which formula is needed.

Is the answer reasonable? Compare the size of the result with a rough estimate made in your head.

Reminder Bring the data sheet and a ruler to the next lab session.

Table 1: Measurements of a falling steel ball

Trial	Height (m)	Time (s)	Speed (m/s)
1	0.50	0.32	3.1
2	1.00	0.45	4.4
3	1.50	0.55	5.4
4	2.00	0.64	6.3
5	2.50	0.71	7.0

A short Python function for this step is shown below.

```
def sieve(n):
    flags = [True] * (n + 1)
    flags[0:2] = [False, False]
    for p in range(2, int(n ** 0.5) + 1):
        if flags[p]:
            for m in range(p * p, n + 1, p):
                flags[m] = False
    return [i for i, ok in enumerate(flags) if ok]
```

A short Python function for this step is shown below.

```
def mean_and_variance(xs):
    n = len(xs)
    mean = sum(xs) / n
    var = sum((x - mean) ** 2 for x in xs) / (n - 1)
    return mean, var
```

Key Takeaways

- A rate of change is a ratio of two differences.
- Neighbouring table rows give the best estimate of a local rate.
- State the unit of every rate you report.

Choosing Between a Linear Model and an Exponential Model for a Small Set of Experimental Data

A straight line and a curve that bends upward can both pass close to a handful of points. The choice should rest on the process that produced the data, not only on how well the curve fits.

Draw a picture first. Even a crude sketch of the situation prevents many sign and direction errors.

State the assumptions clearly. Ignoring air resistance or treating a gas as ideal is acceptable only if it is said aloud.

Convention Angles are measured anticlockwise from the positive horizontal axis.

Example

A solution is made by dissolving 18 g of salt in water to make 250 cm^3. Find the concentration in g per cm^3.

Solution

The concentration is 18 / 250 = 0.072 g/cm^3.

A short Python function for this step is shown below.

```
def fit_line(xs, ys):
    n = len(xs)
    sx, sy = sum(xs), sum(ys)
    sxy = sum(x * y for x, y in zip(xs, ys))
    sxx = sum(x * x for x in xs)
    slope = (n * sxy - sx * sy) / (n * sxx - sx ** 2)
    return slope, (sy - slope * sx) / n
```

A short Python function for this step is shown below.

```
def transpose(m):
    return [list(row) for row in zip(*m)]

a = [[1, 2, 3], [4, 5, 6]]
print(transpose(a))
```

Check Your Understanding

1. Explain in one sentence why the slope of a position graph is a velocity.
2. A graph is flat between t = 2 s and t = 5 s. What does that say about the motion?

A Careful Look at Units, Dimensions and the Common Mistakes Made When Converting Between Them

A numerical answer without a unit is incomplete, and a unit with the wrong dimensions is a sign of an error earlier in the work. Checking both takes only a few seconds.

Start from the definition. When a formula is forgotten, the definition of the quantity usually rebuilds it.

Label every axis and every curve. A graph without labels and units cannot be read by anyone else.

Name: _____

Student number: _____

Date of laboratory: _____

A short Python function for this step is shown below.

```
def binary_search(items, target):
    lo, hi = 0, len(items) - 1
    while lo <= hi:
        mid = (lo + hi) // 2
        if items[mid] == target:
            return mid
        if items[mid] < target:
            lo = mid + 1
        else:
            hi = mid - 1
    return -1
```

A short Python function for this step is shown below.

```
counts = {}
for word in text.split():
    counts[word] = counts.get(word, 0) + 1
top = sorted(counts, key=counts.get, reverse=True)[:5]
```

Key Takeaways

- Area under a velocity graph is displacement.

- Regions below the axis count as negative.
- Check the answer against a rough area estimate.

From Words to Equations: Translating Short Problem Statements into Algebra Without Losing Information

Most algebra mistakes are made before any algebra is done, when a sentence is turned into an equation. Writing down what every symbol stands for removes much of the risk.

One change at a time. Vary a single quantity in each trial so that the effect can be traced to its cause.

Keep a record of failed attempts? They explain later why a particular approach was abandoned.

Table 2: Selected properties of four metals

Metal	Symbol	Density (g/cm³)	Melting point (K)
Aluminium	Al	2.70	933
Copper	Cu	8.96	1358
Iron	Fe	7.87	1811
Zinc	Zn	7.14	693

Example

The sum of two numbers is 25 and their difference is 7. Find both numbers.

Solution

Adding the two equations gives twice the larger number, 32, so the larger number is 16 and the smaller is 9.

A short Python function for this step is shown below.

```
class Vector:
    def __init__(self, x, y):
        self.x, self.y = x, y

    def dot(self, other):
        return self.x * other.x + self.y * other.y
```

A short Python function for this step is shown below.

```
def celsius_to_kelvin(c):
    if c < -273.15:
        raise ValueError("below absolute zero")
    return c + 273.15
```

Check Your Understanding

1. Why is it risky to round 1/3 to 0.33 in the first line of a long calculation?
2. How many significant figures does 0.00450 have?

Why Some Chemical Reactions Reach Equilibrium and What the Equilibrium Constant Tells Us About Them

A reversible reaction does not stop; the forward and reverse rates become equal. The equilibrium constant compares the amounts of products and reactants once that balance has been reached.

Use consistent notation throughout the whole solution. Switching between two symbols for the same quantity is a frequent source of confusion.

Look for a pattern in the first few cases. A conjecture formed from small examples can then be tested on a larger one.

The same idea written in C looks like this.

```
int sum_array(const int *a, int n) {
    int total = 0;
    for (int i = 0; i < n; i++) {
        total += a[i];
    }
    return total;
}
```

The same idea written in C looks like this.

```
void swap(int *p, int *q) {
    int tmp = *p;
    *p = *q;
    *q = tmp;
}
```

Key Takeaways

- Round only at the end of a calculation.
- Keep two spare digits in every intermediate step.
- Report the answer with the precision of the data.

Sketching Polynomial Functions by Finding Intercepts, Turning Points and the Long-Run Behaviour

The shape of a polynomial can be predicted from a few facts: its degree, its roots, and the sign of the leading coefficient. A rough sketch based on these facts is often enough.

Read the question twice. The second reading usually reveals a condition that the first one skipped.

Substitute back to check. Putting the solution into the original equation catches arithmetic slips and extraneous roots.

Student signature	Instructor signature

Example

A data set has the values 4.2, 5.1 and 6.0. Find the mean.

Solution

The mean is (4.2 + 5.1 + 6.0) / 3 = 5.1.

The same idea written in C looks like this.

```
struct point {
    double x;
    double y;
};

double norm2(struct point p) {
    return p.x * p.x + p.y * p.y;
}
```

The same idea written in C looks like this.

```
unsigned long factorial(unsigned n) {
    unsigned long r = 1;
    while (n > 1) {
        r *= n--;
    }
    return r;
}
```

Check Your Understanding

1. Name one situation in which a straight line is the wrong model for data.
2. What does a negative residual tell you about a data point?

Interpreting the Slope and the Intercept of a Line of Best Fit in the Context of an Experiment

The slope of a fitted line is a rate with units, and the intercept is a starting value. Reading both in the language of the experiment turns a formula into a statement about the world.

Estimate the uncertainty. Half of the smallest scale division is a sensible starting point for a single reading.

Take a break when stuck: A short walk often clears the way to an idea that hours at the desk did not.

Table 3: Summary of quiz scores by group

Group	Students	Mean score	Standard deviation
Morning	28	71.4	9.2
Afternoon	31	68.9	10.5
Evening	19	74.2	8.1

The same idea written in C looks like this.

```
#include <stdio.h>

int main(void) {
    for (int k = 1; k <= 5; k++)
        printf("%d squared is %d\n", k, k * k);
    return 0;
}
```

The same idea written in C looks like this.

```
int count_bits(unsigned v) {
    int n = 0;
    for (; v; v >>= 1)
        n += v & 1u;
    return n;
}
```

Key Takeaways

- Choose the model from the process, not from the plot alone.
- Compare residuals, not only the correlation.
- A good fit inside the data says little outside it.

Sampling, Uncertainty and the Difference Between a Mistake in the Method and a Random Fluctuation

No measurement is exact, and repeated readings scatter around a typical value. A systematic offset, by contrast, moves every reading in the same direction.

Explain the result in words. A final sentence in plain language shows that the numbers have been understood.

Mind the difference between mass and weight. Mass is measured in kilograms and weight, a force, in newtons.

Example

A sample of 80 g of a radioactive isotope has a half-life of 6 hours. How much remains after 18 hours?

Solution

Eighteen hours is 3 half-lives, so the mass is 80 / 2^3 = 10 g.

The same idea written in C looks like this.

```
double mean(const double *x, size_t n) {
    double s = 0.0;
    for (size_t i = 0; i < n; i++)
        s += x[i];
    return s / n;
}
```

From the command line the steps are as follows.

```
for f in data/*.csv; do
    echo "$f has $(wc -l < "$f") lines"
done
```

Check Your Understanding

1. A student writes 5 m + 3 s. What is wrong with this expression?
2. Convert 72 km/h to metres per second.

Counting Techniques for Arrangements and Selections, with Worked Examples from Games of Chance

Counting problems become easier once it is clear whether order matters and whether repetition is allowed. The two questions fix which formula applies.

Collect like terms before factoring. Neater expressions have fewer chances for sign errors.

Compare with a limiting case. Setting a parameter to zero or letting it grow often gives a result that is known already.

Given. The mass is 2.0 kg, the height is 5.0 m and the starting speed is zero.

Find. The speed of the mass just before it reaches the ground.

From the command line the steps are as follows.

```
mkdir -p results/plots
python analyse.py --input raw.csv --out results/
ls -l results
```

From the command line the steps are as follows.

```
cut -d, -f2 grades.csv | sort -n | uniq -c | sort -rn | head
```

Key Takeaways

- Every equation must balance in its units.
- Convert before you substitute.
- Write the unit on every line of the work.

Solving Systems of Linear Equations by Elimination and Recognising When No Solution Exists at All

Elimination replaces a pair of equations by an equivalent pair in which one unknown has disappeared. A row that reduces to zero equals a nonzero number reveals that the system has no solution.

Report the final answer with its unit. A number alone is not an answer in any science course.

Share the work with a partner? Explaining a step aloud is the quickest way to find a gap in it.

Table 4: Half-lives of some radioactive isotopes

Isotope	Protons	Neutrons	Half-life
^{14}C	6	8	5730 years
^{60}Co	27	33	5.27 years
^{131}I	53	78	8.0 days
^{238}U	92	146	4.5 billion years

Example

A rectangle has a perimeter of 30 cm and one side of 4 cm. Find its area.

Solution

The other side is 30/2 - 4 = 11 cm, so the area is 44 cm^2.

From the command line the steps are as follows.

```
find . -name "*.log" -mtime +30 -print
grep -c "ERROR" run.log
```

From the command line the steps are as follows.

```
tar -czf backup.tar.gz notes/ figures/
tar -tzf backup.tar.gz | head -n 5
```

Check Your Understanding

1. State two assumptions behind the ideal gas law.

2. What quantity stays constant at equilibrium: the amounts or the rates?

Writing a Clear Laboratory Report That Another Student Could Follow Without Asking Any Questions

A good report states the question, the method, the data and the conclusion, in that order. Each part should be short enough to read in a minute.

From the command line the steps are as follows.

```
name="lab3"
echo "Preparing $name"
cp template.tex "$name.tex"
head -n 3 "$name.tex"
```

A database query for the same summary is given next.

```
SELECT student, AVG(score) AS mean_score
FROM quiz
GROUP BY student
HAVING COUNT(*) >= 3
ORDER BY mean_score DESC;
```

Checking an Answer for Reasonableness Before Submitting It: Estimates, Limiting Cases and Symmetry

Before an answer is handed in, a quick estimate, a limiting case or a symmetry argument can reveal a slip. Such checks are cheap and surprisingly effective.

A database query for the same summary is given next.

```
SELECT s.name, c.title
FROM enrolment AS e
JOIN student AS s ON s.id = e.student_id
JOIN course AS c ON c.id = e.course_id;
```

In JavaScript the idea is expressed like this.

```
function clamp(x, lo, hi) {
  return Math.min(hi, Math.max(lo, x));
}

console.log(clamp(12, 0, 10));
```

Key Terms

Derivative The limit of the ratio of the change in a function to the change in its input; it gives the slope of the tangent line.

Integral A limit of sums that measures accumulated change, such as the area under a curve.

Limit The value that a function approaches as its input gets arbitrarily close to a chosen point.

Matrix A rectangular array of numbers arranged in rows and columns.

Determinant A single number computed from a square matrix that tells whether the matrix can be inverted.

Eigenvalue A scalar by which a matrix stretches a special direction called an eigenvector.

Vector A quantity with both a size and a direction, often written as a column of numbers.

Scalar A quantity described by a single number and, in physics, a unit.

Domain The set of all inputs for which a function is defined.

Range The set of all outputs that a function actually produces.

Asymptote A line that a curve approaches without ever reaching it.

Polynomial A sum of terms, each a constant times a whole-number power of a variable.

Logarithm The exponent to which a base must be raised to give a stated number.

Sequence An ordered list of numbers, usually given by a rule.

Series The sum of the terms of a sequence.

Probability A number between zero and one that measures how likely an event is.

Mean The sum of a set of values divided by how many values there are.

Variance The average squared distance of the values in a data set from their mean.

Hypothesis A statement about the world that an experiment is designed to test.

Isotope One of several forms of an element that have the same number of protons but different numbers of neutrons.

Ion An atom or molecule that carries a net electric charge.

Mole The amount of substance that contains as many particles as there are atoms in twelve grams of carbon-12.

Molarity The number of moles of solute in one litre of solution.

Catalyst A substance that speeds up a reaction without being used up.

Equilibrium A state in which the forward and reverse rates of a process are equal and the amounts stop changing.

Half-life The time needed for half of the nuclei in a radioactive sample to decay.

Velocity The rate of change of position, with a direction.

Acceleration The rate of change of velocity.

Momentum The product of an object's mass and its velocity.

Frequency The number of complete cycles of a wave that pass a point each second.

Wavelength The distance between two matching points on neighbouring cycles of a wave.

Amplitude The largest displacement of a wave from its rest position.

Algorithm A finite list of precise steps that solves a class of problems.

Variable A name that stands for a value which may change.